

Affine Gaussian score transfer, predictable-feedback divergence, polynomial-response support, and aggregate collar boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1. Purpose and scope

R-132 selected Gaussian score transfer and a complete $\gamma = 4$ shell estimate as its apparent successor. R-133 audits both suggestions against the complete owner and changes the route in two precise ways.

1. For a genuinely affine conditional Gaussian future block, the complete current-trace response has an exact common-heat Hermite-score formula. It removes every derivative of C_6 , including differentiated floor denominators, while retaining the raw signed six-row Pauli-Fierz kernel.
2. A general predictable feedback chart is not affine in the Gaussian sources. Its exact score contains an inverse feedback Jacobian, a Gaussian double divergence, and the R-128 connection term. The declared finite-energy control class does not control those quantities.
3. Common heat scores only the replica-synchronous direction. The replica-relative derivative remains explicit and carries exactly one half of the conditional variation variance.
4. R-083 kills the stopped polynomial current, not the complete polynomial physical response. A response-level zero theorem is conditional on a root intertwiner and covariance/forest matching and uses the safe collar $C = 4$, not the current-level collar $C = 3$.
5. The full joint rational row has a nonzero boundary layer. A scalar sharp-Fourier surrogate for $\mathcal{B}_4^{\text{out}}$ grows like $b^7 e^{-3/2}$, whereas the once-owned sextic grows like b^6 . Therefore the existing sixth-amplitude deterministic/current-coefficient route cannot prove $\gamma = 4$.
6. A $\gamma = 4$ theorem is not logically necessary for fixed-collar three-channel acceptance. Any proved aggregate tail with $\gamma > 0$ can be made smaller than a fixed strict headroom. In particular, a still-unproved one-use bound for R-091's accepted $\gamma = 7/12$ ledger would suffice after an explicit collar choice, provided the non-tail contribution remains below the exact shifted threshold.

The result is T4 at fixed finite cutoff and positive density floor. It does not prove the complete production score-weighted owner estimate, a one-use $\mathcal{B}_{7/12}^{\text{out}}$ bound, production C_{mix} , C_{far} , c_{bal} , low $D/K/R_0/S_0$, matching energy, a strict augmented gap, the absolute anchor, $\text{OVERLAP}_{\text{src}}$, Nelson, removals, an interacting measure, or Sector-A closure.

2. Owner and conditioning firewall

Fix one finite temporally faithful chart, one positive floor, one root after all representation-preserving same-root visits have been recombined, one member of an exhaustive deterministic output family, and one spatial derivative direction. Every use of “complete” below retains the entire contraction-closed cluster

sum before shelling. The heat variable r is common to both future replicas and is integrated once outside the replica expectation. It is not independently replicated. The trace probe is independent and once owned.

R-132 identifies the direct physical current–trace response as

$$q_{\text{comp}} = -\frac{1}{2}D^2\mathcal{T} = \frac{1}{2}D^2(\mathcal{F}_{063}^{\text{ad}} - \mathcal{V}_{\text{fut}}). \quad (2.1)$$

The R-063 forest and future variance are alternate coordinates for this same owner. Low, paid, rational, Cartan, heat, trace, and forest pieces are not added again after (2.1). The fixed-chart action Hessian is

$$D^2\mathcal{A}_J(h)[u, v] = \frac{9}{10}\langle u, v \rangle + q_{\text{comp}}(L_\pi u, L_\pi v) + q_6(Z_h; L_\pi u, L_\pi v). \quad (2.2)$$

The source term stays outside the physical response and is used once.

3. Exact affine common-heat Gaussian score theorem

Condition on the strict past and retained root. First fix one owner tuple $(k, \mathcal{C}, i)$ and suppress its indices. Let its complete future block be affine,

$$W_{f,r} = \overline{W}_r + S\xi_f, \quad \xi_f, \xi_{f'} \stackrel{\text{iid}}{\sim} N(0, I), \quad (3.1)$$

and put

$$X_f = P_{\mathcal{C}}C_6(W_{f,r})\partial_i W_{f,r}, \quad K = \text{Re}\langle X_f, X_{f'} \rangle. \quad (3.2)$$

For the independent once-owned trace probe G'_k , put

$$T = \mathbb{E}_g \|P_{\mathcal{C}}C_6(W_{f,r})\partial_i G'_{k,g}\|^2. \quad (3.3)$$

Let the physical variations be genuine Gaussian mean shifts

$$z = Sa, \quad w = Sb. \quad (3.4)$$

Define the one- and two-copy Hermite scores

$$H_{a,b}^{(1)}(\xi) = (a \cdot \xi)(b \cdot \xi) - a \cdot b, \quad (3.5)$$

$$H_{a,b}^{(2)}(\xi, \xi') = [a \cdot (\xi + \xi')][b \cdot (\xi + \xi')] - 2a \cdot b. \quad (3.6)$$

Theorem 3.1 (affine mixed-replica score identity). Under the firewall above, suppose first that the same conditional Gaussian source coordinate realizes all owner tuples and that a, b are its common shift directions. Then

$$\boxed{q_{\text{comp}}(Sa, Sb) = \frac{1}{2} \sum_{k, \mathcal{C}, i} \mathbb{E}_r \mathbb{E} \left[H_{a,b}^{(2)} K - H_{a,b}^{(1)} T \right]}. \quad (3.7)$$

Proof. Both future replicas receive the same mean shift. For a two-parameter shift $h = ta + ub$, the product Gaussian density ratio is

$$\exp\{h \cdot (\xi + \xi') - |h|^2\}, \quad (3.8)$$

whose mixed second derivative in directions a, b is (3.6). The one-copy density ratio has second score (3.5). Apply these two identities to $\frac{1}{2}\mathbb{E}_{f,f'}K - \frac{1}{2}\mathbb{E}_fT$, keeping r fixed, and then integrate the common heat once. This is exactly (2.1). $\square$

If different owner tuples use different affine source coordinates, the single-owner identity preceding the sum remains exact with indexed $S_{k\mathcal{C}i}, a_{k\mathcal{C}i}, b_{k\mathcal{C}i}$; (3.7) then means the sum of those ownerwise identities. It does not identify distinct conditional laws or replace the exhaustive cluster sum by one selected output.

No DC_6 or D^2C_6 occurs in (3.7). The raw signed kernel remains

$$c_0\mathbf{J}\mathbf{J} + c_1\{\mathbf{J}\mathbf{J} - \alpha\tau'\mathbf{J}\mathbf{M} - \alpha\tau\mathbf{M}\mathbf{J} + \alpha^2\tau\tau'\mathbf{M}\mathbf{M}\}. \quad (3.9)$$

Thus the denominators in τ, τ' are not cancelled, but they are not differentiated. The complete row sum has the raw floor-independent envelope

$$\|C_6(W)V\|^2 \leq \beta_{\text{op}}|W|^2|V|^2, \quad \beta_{\text{op}} = 4(c_0 + c_1) = \frac{339}{2000P}. \quad (3.10)$$

This envelope belongs to the complete row expression. Separating τ before applying it can recreate false inverse-floor losses. The live 2680/729 Cartan owner remains encoded in (3.7).

4. Exact covariance cost

Let a physical affine Gaussian have covariance Γ . For $z, w \in \text{Ran } \Gamma^{1/2}$ define

$$\kappa(z, w) = \langle z, \Gamma^\dagger w \rangle, \quad D_\Gamma(z, w) = \kappa(z, z)\kappa(w, w) + \kappa(z, w)^2. \quad (4.1)$$

In physical variables the scores are

$$s_z = \langle \Gamma^\dagger z, W - \overline{W} \rangle, \quad (4.2)$$

$$H_{z,w}^{(1)} = s_z s_w - \kappa(z, w), \quad (4.3)$$

$$H_{z,w}^{(2)} = (s_z + s'_z)(s_w + s'_w) - 2\kappa(z, w). \quad (4.4)$$

Wick's formula gives the exact costs

$$\boxed{\|H_{z,w}^{(1)}\|_2^2 = D_\Gamma(z, w), \quad \|H_{z,w}^{(2)}\|_2^2 = 4D_\Gamma(z, w).} \quad (4.5)$$

Consequently, in the common-coordinate setting of Theorem 3.1,

$$\boxed{|q_{\text{comp}}(z, w)| \leq D_\Gamma(z, w)^{1/2} \sum_{k, \mathcal{C}, i} \mathbb{E}_r \left(\|K\|_2 + \frac{1}{2} \|T\|_2 \right).} \quad (4.6)$$

For owner-dependent affine coordinates the precise replacement is the ownerwise sum

$$\sum_{k, \mathcal{C}, i} \mathbb{E}_r \left[D_{\Gamma_{k\mathcal{C}i, r}}(z, w)^{1/2} \left(\|K\|_2 + \frac{1}{2} \|T\|_2 \right) \right], \quad (4.6a)$$

with the physical directions and covariance interpreted in that owner's coordinate. No common covariance may be pulled outside this sum unless it is actually shared by all owners.

The range condition is load-bearing. Outside $\text{Ran } \Gamma^{1/2}$ the translated and original Gaussian laws are singular. Even inside the range, for

$$\Gamma_\varepsilon = \text{diag}(1, \varepsilon^2), \quad z = e_2, \quad (4.7)$$

one has $\kappa(z, z) = \varepsilon^{-2}$. No physical L^2 estimate hides this inverse. In source coordinates $z = Sa$, the minimum cost is the square of the projection of a onto $(\ker S)^\perp$.

5. Common and relative replica directions

The affine theorem has the same direction on both replicas. General feedback does not. The following identity locates the difference exactly. Let $A \sim N(0, \Sigma)$ be common heat and let $\Phi(A + x, A + y)$ be smooth. For two replica directions p, q , write

$$c = \frac{p+q}{2}, \quad d = \frac{p-q}{2}, \quad \nabla_- := \nabla_x - \nabla_y. \quad (5.1)$$

Then

$$p \cdot \nabla_x + q \cdot \nabla_y = c \cdot \nabla_A + d \cdot \nabla_-. \quad (5.2)$$

For two direction pairs $(p_z, q_z), (p_w, q_w)$, Gaussian integration by parts gives

$$\begin{aligned} \mathbb{E}_A L_z L_w \Phi &= \mathbb{E}_A [H_\Sigma(c_z, c_w; A) \Phi] \\ &\quad + \mathbb{E}_A [S_\Sigma(c_z; A) d_w \cdot \nabla_- \Phi] \\ &\quad + \mathbb{E}_A [S_\Sigma(c_w; A) d_z \cdot \nabla_- \Phi] \\ &\quad + \mathbb{E}_A [(d_z \cdot \nabla_-)(d_w \cdot \nabla_-) \Phi] \end{aligned} \quad (5.3)$$

where

$$S_\Sigma(c; A) = \langle \Sigma^\dagger c, A \rangle, \quad (5.4)$$

$$H_\Sigma(c, w; A) = S_\Sigma(c; A) S_\Sigma(w; A) - \langle c, \Sigma^\dagger w \rangle. \quad (5.5)$$

For iid conditional replica directions p, p' in any Hilbert metric,

$$\mathbb{E} \left\| \frac{p+p'}{2} \right\|^2 = \|\mathbb{E} p\|^2 + \frac{1}{2} \text{Var}(p), \quad (5.6)$$

$$\mathbb{E} \left\| \frac{p-p'}{2} \right\|^2 = \frac{1}{2} \text{Var}(p), \quad \mathbb{E} \langle p+p', p-p' \rangle = 0. \quad (5.7)$$

Thus common heat transfers the synchronous direction. The unsmoothed relative derivative is localized to one half of the conditional variation variance. In the affine mean-shift theorem $p = p'$ and it vanishes. In a predictable feedback chart it is an additional owner, not a free cancellation.

6. Exact predictable-feedback divergence theorem

On a smooth finite triangular source chart write

$$Z_h(\xi) = L_\pi \{\xi + h(\xi)\}, \quad J_h = I + D_\xi h. \quad (6.1)$$

For legal fixed-law control directions u, v , put

$$a = J_h^{-1} u, \quad b = J_h^{-1} v, \quad d = J_h^{-1} D_\xi^2 h[a, b]. \quad (6.2)$$

For standard Gaussian divergence define

$$\delta(a) = \xi \cdot a - \text{div } a, \quad (6.3)$$

and for a matrix field M define

$$\delta^2(M) = \xi^T M \xi - \text{Tr } M - 2\xi \cdot \text{div } M + \text{div}^2 M. \quad (6.4)$$

Theorem 6.1 (feedback-corrected score). Assume that J_h is pointwise invertible and that h, u, v, F , together with the fields in (6.2)–(6.4), lie in a Gaussian–Sobolev/integrability core on which both adjunctions below are valid. Then, for every such smooth cylindrical scalar owner F ,

$$\boxed{\mathbb{E} D^2 F(Z_h)[L_\pi u, L_\pi v] = \mathbb{E} F(Z_h) \left[\delta^2(\text{sym}(a \otimes b)) - \delta(d) \right]}. \quad (6.5)$$

Proof. Since $J_h a = u$ and $J_h b = v$,

$$D_\xi^2(F \circ Z_h)[a, b] = D^2 F(Z_h)[L_\pi u, L_\pi v] + DF(Z_h)[L_\pi D^2 h[a, b]]. \quad (6.6)$$

The definition of d gives

$$D_\xi(F \circ Z_h)[d] = DF(Z_h)[L_\pi D^2 h[a, b]]. \quad (6.7)$$

Apply Gaussian adjunction twice to (6.6) and once to (6.7). $\square$

For the replica kernel, apply (6.5) on the product Gaussian space. For the trace use one copy. Thus (3.7) remains exact after replacing $H^{(2)}$ and $H^{(1)}$ by the corresponding product- and single-space scores

$$\mathfrak{S}_{u,v} = \delta^2(\text{sym}(a \otimes b)) - \delta(d). \quad (6.8)$$

The coefficient derivatives disappear, but the price is J_h^{-1} , derivatives of a, b inside δ^2 , and the feedback connection d . After expansion, up to third Gaussian derivatives of h and second Gaussian derivatives of u, v can occur. These are not among the declared finite-energy graph coordinates.

The connection term is not cosmetic. On the exact triangular chart

$$Z_h(x, y) = (x, y + \lambda x^2), \quad u = v = e_1, \quad (6.9)$$

the owner $F(z) = z_2^2/2$ has physical Hessian zero. The naked double divergence contributes $2\lambda^2$, and $-\delta(d)$ cancels exactly the same amount. The primary and independent executables reconstruct this identity by separate exact polynomial algebras.

7. Finite-energy score-transfer no-go and sufficient gate

Let X, Y be independent standard Gaussians and take the bounded smooth strict-triangular feedback

$$h_N(X, Y) = (0, a \tanh(NX)), \quad 0 < a < 1. \quad (7.1)$$

Its amplitude and every fixed terminal polynomial moment are uniformly bounded. Nevertheless

$$J_{h_N}^{-1} e_1 = (1, -aN \text{sech}^2(NX)), \quad (7.2)$$

and for $N \geq 1$,

$$\boxed{\mathbb{E} \|J_{h_N}^{-1} e_1\|^2 \geq 1 + 2a^2 \varphi(1) \text{sech}^4(1)N.} \quad (7.3)$$

Indeed restrict to $|X| \leq N^{-1}$, where the Gaussian density is at least $\varphi(1)$ and $\text{sech}^4(NX)$ is at least $\text{sech}^4(1)$. Thus bounded predictable source amplitude, triangularity, and determinant one do not control the score norm. This is a method no-go for global source translation, not a counterexample to the physical fixed-law Hessian.

The exact remaining sufficient estimate is now visible. Since

$$q_6(Z; H, H) \geq \frac{9}{10} \mathbb{E} \int |Z|^4 |H|^2, \quad (7.4)$$

a production theorem would follow if the complete signed score-weighted raw owner satisfied

$$\frac{1}{2} \sum \left(\|\mathfrak{S}_{u,u}^{(2)}\|_2 \|K\|_2 + \|\mathfrak{S}_{u,u}^{(1)}\|_2 \|T\|_2 \right) \leq a_0 \|u\|^2 + b_0 \mathbb{E} \int |Z|^4 |L_\pi u|^2 \quad (7.5)$$

with

$$a_0 < \frac{9}{10}, \quad b_0 < \frac{9}{10}. \quad (7.6)$$

R-132 proves such a margin only on its standard-Gaussian scalar ray. No complete production theorem presently supplies (7.5).

8. Polynomial response support and the noncommuting square

R-083 proves that a stopped polynomial current is zero beyond its safe collar $C = 3$. The complete response contains two additional coordinates. Restricted to the three linear Pauli–Fierz rows, the owner bookkeeping is

$$q_{\text{poly}} = q_{\text{R083 stop}} + q_{\Sigma,+} - \frac{1}{2} D^2 (\Delta \mathcal{V}_{\text{fut}}^L - \Delta \mathcal{F}_{063}^{L,\text{ad}}). \quad (8.1)$$

Here $q_{\Sigma,+} \succeq 0$ is the common-heat Gram lift. The last term is the exact R-125 future-variance/forest residual. R-083 kills only the first term.

Theorem 8.1 (conditional stopped-polynomial response zero). Assume all of the following after same-root visit recombination.

1. Differentiation is at fixed law, cutoff, floor, heat, covariance owner, filtration, and output projections.
2. The aggregate is a genuine R-083 predictable changing-current tower with the same moving endpoint and both predictable drifts.
3. Endpoints, first and second physical variations, drift fields, and owned derivative probes through the stop n are supported in Q_n .
4. Common heat is centered, deterministic or strict-past/fresh-root independent, and spatially stationary.
5. Output projections are exhaustive, deterministic, mutually orthogonal, commute with the physical shell cut, and are summed in contraction-closed clusters before shelling.
6. The shellwise linear covariance/forest residual in (8.1) vanishes; equivalently one may impose the sufficient matching conditions $K_j = 0$, $\mu = 0$, $V = \Gamma_{\text{own}}$ with no complement leakage.

Then the complete paired polynomial-response symbol is supported in Q_{n+2} and

$$\boxed{\Pi_m Q_{\text{poly},n} = 0 \quad (m \geq n + 4).} \quad (8.2)$$

Consequently the conditional polynomial far constant is zero for $C \geq 4$.

Proof. R-083 gives the current tower and cancels the matched residual. A polynomial current is quadratic, so it is supported in Q_{n+1} . The paired response contracts two current/variation factors and is supported in Q_{n+2} . The safe projector convention places its threshold one further dyadic step away, giving $m \geq n + 4$. $\square$

The offsets are therefore

$$C_{\text{current}} = 3, \quad C_{\text{response}} = 4, \quad C_{\text{Cartan principal}} = 5. \quad (8.3)$$

They must not be interchanged.

Neither the root intertwiner nor the shellwise covariance/forest matching is known in production. The exact common-heat fixture gives

$$q_{\Sigma,+} = 2(c_0 + c_1) = \frac{339}{4000P} > 0, \quad (8.4)$$

even though averaging the heat before squaring gives zero. Independently, R-083's adapted Hermite fixture gives

$$\mathbb{E}\mathcal{W} = -8(c_0 + c_1)\lambda^2 = -\frac{339}{1000P}\lambda^2 < 0. \quad (8.5)$$

Thus there is no literal full-owner-to-stopped-current equality. Equations (8.4)–(8.5) are method boundaries, not a production FAR counterexample.

9. Joint rational boundary layer and the $\gamma = 4$ no-go

On the one-real Pauli slice let

$$u = se_1, \quad v = pe_1, \quad \chi = w = 0, \quad (9.1)$$

and define

$$f_e(s) = s - \frac{5}{9} \frac{s^3}{s^2 + e}. \quad (9.2)$$

The complete diagonal six-row Gram is

$$G = 4p^2\{c_0s^2 + c_1f_e(s)^2\}. \quad (9.3)$$

This is the joint row sum; no JJ, JM, MJ, MM piece has been bounded separately. After removing its polynomial and constant principal pieces,

$$R_e(s) := f_e(s)^2 - \frac{16}{81}s^2 - \frac{40}{81}e = eH(s/\sqrt{e}), \quad (9.4)$$

where

$$\boxed{H(z) = -\frac{5}{27(1+z^2)} - \frac{25}{81(1+z^2)^2}.} \quad (9.5)$$

Thus joint cancellation improves but does not remove the floor boundary layer.

Take $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ with

$$\widehat{g}(k) = \frac{1}{2\pi} \int_0^{2\pi} g(x)e^{-ikx} dx, \quad \|g\|_2^2 = 2\pi \sum_{k \in \mathbb{Z}} |\widehat{g}(k)|^2. \quad (9.6)$$

For $N = 2^j$, set

$$s(x) = b \cos(Nx), \quad p(x) = \sin(Nx). \quad (9.7)$$

Put $\delta = \sqrt{e}/b$, $\kappa = \operatorname{arsinh} \delta$, and $c = \sqrt{1 + \delta^2}$. For $n \geq 2$, the exact coefficient at mode $2n$ of $4c_1 R_e(b \cos x)$ is

$$a_n = 4c_1 e(-1)^{n+1} e^{-2n\kappa} \left\{ \frac{5\delta}{27c} + \frac{25\delta^2}{81c^2} (n + \coth(2\kappa)) \right\}. \quad (9.8)$$

Multiplication by $\sin^2 x$ gives

$$g_n = \frac{1}{2}a_n - \frac{1}{4}(a_{n-1} + a_{n+1}), \quad (9.9)$$

the full Gram coefficient at mode $2n$ once $n \geq 3$, because every polynomial piece ends at mode four. Adjacent a_n have opposite signs, so $(-1)^{n+1}g_n > 0$; every far even harmonic survives. With $y = Nx$, the sharp region $|k| > 16N$ is $n \geq 9$, and Parseval gives the exact identity

$$N^{-8} \|\partial_x^4 P_{|k| > 16N} G\|_2^2 = 4\pi \sum_{n \geq 9} (2n)^8 |g_n|^2. \quad (9.10)$$

There are two zeros of $\cos y$ per carrier period, hence $2N$ zeros on $\mathbb{T}$; the change $y = Nx$ accounts for all of them. Blowing up those zeros gives the exact line integral

$$\int_{\mathbb{R}} |H^{(4)}(z)|^2 dz = \frac{2062375\pi}{23328}. \quad (9.11)$$

It follows that

$$\lim_{b \rightarrow \infty} \frac{e^{3/2}}{b^7} N^{-8} \|\partial_x^4 P_{|k| > 16N} G\|_2^2 = \frac{2062375\pi}{729} c_1^2. \quad (9.12)$$

The independent executable evaluates the exact series (9.10) at $e = 1$, $b = 20, 50, 100$; after division by the right asymptotic scale its ratios are $1.00438708256\dots$, $1.00070316267\dots$, and $1.00017576283\dots$

Define the R-091-normalized sharp surrogate

$$\mathfrak{B}_4^\# := 2^{-40} N^{-8} \|\partial_x^4 P_{|k| > 16N} G\|_2^2. \quad (9.13)$$

The factor 2^{-40} is the exact shift from $m - j$ to $m - j - 5$ at $\gamma = 4$. A smooth Littlewood–Paley ledger is comparable to (9.13) with partition constants, but is not literally identical to it.

For a generous comparison with the available once-owned sixth-amplitude majorant, use the sextic potential value on the state profile (not the terminal sextic Hessian in direction p):

$$\frac{3}{20} \int_0^{2\pi} s^6 dx = \frac{3\pi}{32} b^6. \quad (9.14)$$

Therefore

$$\frac{2^{40} \mathfrak{B}_4^\#}{(3/20) \int s^6} \sim \frac{445473}{16000P^2} \frac{b}{e^{3/2}}. \quad (9.15)$$

At the production $P = 4 + 10^{-12}$, the derived coefficient is $1.740128906249\dots$. The ratio is unbounded even at fixed positive e . More generally the boundary layer scales as

$$N^{-2\gamma} \| |D|^\gamma P_{\text{far}} G \|_2^2 \asymp b^{2\gamma-1} e^{5/2-\gamma}. \quad (9.16)$$

A sixth-amplitude budget can cover at most $\gamma \leq 7/2$; $\gamma = 4$ requires a seventh amplitude moment.

This fires a precisely scoped method no-go. It rejects deriving $\gamma = 4$ from the complete Pauli–Fierz coefficient algebra plus the existing sixth-amplitude/ Z^6 majorants. It is not a literal counterexample to $\mathcal{B}_4^{\text{out}}$, which also contains the OU operator, probability-root difference, output multiplier, expectations, root sum, trace, forest, and the production conditional law. A complete signed production cancellation remains possible.

No Gaussian-smoothing conclusion is used here. Convolution in the scalar state variable s is not the production OU/common-heat operation, so it cannot be invoked as a repair without a separate owner-preserving theorem.

10. Exact aggregate collar-to-gap theorem

R-131 writes the remaining source and sextic diagonals as $e, f > 0$, the full cross norm as a , a low block as $D \succeq dI$, and its coupling norm as k . Put

$$\sigma = \frac{k^2}{d}. \quad (10.1)$$

Assume $\sigma < \min(e, f)$. The exact two-channel lower margin is

$$m_2(a) = \frac{e + f - \sqrt{(e - f)^2 + a^2}}{2}. \quad (10.2)$$

Theorem 10.1 (collar headroom criterion). The strict low-coupled condition $k^2 < dm_2(a)$ is equivalent to

$$\boxed{a < a_* := 2\sqrt{(e - \sigma)(f - \sigma)}}. \quad (10.3)$$

Proof. The condition is $m_2(a) > \sigma$. Since $\sigma < \min(e, f)$, both sides remain positive while squaring. The exact factorization is

$$(e + f - 2\sigma)^2 - (e - f)^2 = 4(e - \sigma)(f - \sigma). \quad (10.4)$$

This proves (10.3). $\square$

Suppose a complete aggregate shell theorem supplies, for integer $C \geq 5$,

$$a(C) \leq a_0(C) + M2^{-\gamma(C-5)}, \quad \gamma > 0. \quad (10.5)$$

Then strict acceptance at collar C follows from

$$a_0(C) + M2^{-\gamma(C-5)} < a_*. \quad (10.6)$$

If $a_0(C) \leq a_0 < a_*$ uniformly and $h_* = a_* - a_0$, an integer collar satisfying

$$C > 5 + \frac{1}{\gamma} \log_2 \frac{M}{h_*} \quad (10.7)$$

closes the strict matrix gap. If e, f, d, k vary with the collar, a_* here must be replaced by a strictly positive collar-uniform lower bound. For R-091's accepted $\gamma = 7/12$,

$$C > 5 + \frac{12}{7} \log_2 \frac{M}{h_*}. \quad (10.8)$$

If the tail comes from

$$\mathcal{H}_C(C) \leq 2^{-2\gamma(C-5)} \mathcal{B}_\gamma^{\text{out}}, \quad (10.9)$$

then one may take $M = \sqrt{\mathcal{B}_\gamma^{\text{out}}}$ after the exact operator and owner conversion has been proved.

This is a route correction, not closure. R-091 does not prove the required one-use bound for $\mathcal{B}_{7/12}^{\text{out}}$, and $a_0(C)$ may depend adversely on the collar through the near/balanced band. Moreover the aggregate tail cannot be relabelled as the old uniform 2^{-4C} or 2^{-2C} constants:

$$M2^{-7(C-5)/12} 2^{4C} = M2^{(41/12)C+35/12}, \quad (10.10)$$

$$M2^{-7(C-5)/12} 2^{2C} = M2^{(17/12)C+35/12}. \quad (10.11)$$

Both grow with C . The new route is a fixed-collar aggregate acceptance test, not a proof of the old entrywise exponents. Summing disjoint collar amplitudes costs the finite geometric factor

$$(1 - 2^{-7/12})^{-1}. \quad (10.12)$$

11. Proof-search evidence map

The append-only ledger records EXP-000504–EXP-000513. The compact map is:

Route	Verdict	Exact boundary or next condition
Affine Gaussian score	advanced	Exact complete current–trace identity; raw signed rows and common heat retained.
Score covariance cost	advanced conditionally	Requires Cameron–Martin range and the covariance pseudoinverse.
Common/relative replicas	advanced	Common heat scores the synchronous part; relative variance remains by (5.7).
Predictable score transfer	advanced on core	Exact double divergence and connection; finite-energy extension is unavailable.
Global source translation	failed	Bounded tanh feedback has divergent inverse-Jacobian score cost.
R-083 to physical response	corrected	Conditional zero uses $C = 4$ and requires root intertwining plus covariance/forest matching.
$\gamma = 4$ coefficient route	failed	Joint rational boundary layer needs seventh amplitude moment.
Aggregate $\gamma = 7/12$ collar	advanced conditionally	Any positive gamma can fit a fixed strict headroom; one-use ledger and balanced collar remain open.
Owner synthesis	inconclusive	Production score ledger, low data, matching energy, and absolute anchor remain missing.

12. Devil’s-advocate review

Objection 1: Theorem 3.1 is the production Hessian theorem. UPHELD. It is exact only for an affine conditional Gaussian mean shift. Theorem 6.1 shows the additional inverse-Jacobian, divergence, and connection costs for general predictable feedback.

Objection 2: Gaussian score transfer cancels the rational floor. UPHELD. It removes derivatives of the floor coefficient. The raw τ, τ' denominators and the signed rational/Cartan owner remain inside the score expectation.

Objection 3: Triangularity controls the inverse Jacobian. UPHELD. Triangularity gives determinant one and pointwise invertibility. The bounded tanh family (7.1) makes the relevant inverse norm grow linearly in N .

Objection 4: R-083 already gives zero polynomial physical response at $C = 3$. UPHELD. It gives zero stopped current. The paired response has one additional product support step, common-heat Gram, and the covariance/forest residual. The conditional response theorem uses $C = 4$ and stronger hypotheses.

Objection 5: The boundary-layer family disproves production. VALID WITH MITIGATION. It disproves the deterministic/current-coefficient sixth-amplitude route to $\gamma = 4$. It does not instantiate every OU, trace, forest, conditional-law, and root owner in literal $\mathcal{B}_4^{\text{out}}$.

Objection 6: $\gamma = 7/12$ now closes the shell theorem. UPHELD. Theorem 10.1 is conditional on a one-use bound for $\mathcal{B}_{7/12}^{\text{out}}$ and on collar-uniform non-tail headroom. Both remain open.

Objection 7: The old C_{far} may be reported after choosing one large collar. **UPHELD.** Equations (10.10)–(10.11) show that the corresponding effective old constants grow with the collar. Only the aggregate fixed-collar residual is justified.

13. Result footer

Result ID: A13-CLASSII-AFFINE-GAUSSIAN-SCORE-FEEDBACK-COLLAR-BOUNDARY (R-133).
Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION.
Precise statement: The complete current--trace owner has an exact derivative-free common-heat Hermite-score representation on affine conditional Gaussian future blocks. General predictable charts have the exact corrected score $\delta^2(\text{sym}(a \text{ tensor } b)) - \delta(d)$, whose inverse-Jacobian/divergence cost is not controlled by finite source energy. R-083 gives a conditional polynomial physical-response zero only at safe collar $C=4$ with root-intertwining and covariance/forest matching. The complete rational diagonal boundary layer rejects the existing sixth-amplitude $\gamma=4$ route. Any positive aggregate γ , including the accepted $7/12$, is conditionally sufficient for a fixed strict collar headroom.
Scope: fixed finite cutoff and positive floor; affine conditional Gaussian blocks; smooth finite triangular cylindrical charts; conditional polynomial response theorem; scalar sharp-Fourier rational surrogate; exact deterministic three-channel collar algebra.
Dependencies: A1; R-083, R-087, R-091, R-104, R-109, R-123, R-125, R-128, R-129, R-130, R-131, and R-132.
Evidence grade: ANALYTIC, EXACT, EXECUTED.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/a13_classii_affine_gaussian_score_feedback_collar_boundary_verify.py
Expected output: R-133 integrated PASS; primary 42/42 and non-importing standard-library independent 38/38 children PASS; PDF and generated-surface gates PASS.
Falsification gate: failure of either Hermite score identity or L2 cost; common/relative decomposition; feedback connection identity; bounded-tanh inverse-cost lower bound; support offsets $3/4/5$; common-heat or adapted-covariance fixture; rational remainder; H-four integral, exact periodic far-series normalization, or asymptotic; sixth-versus-seventh amplitude boundary; collar factorization; any pinned authority, executable, JSON, exploration, negative/audit, PDF, manifest, or public-surface assertion.
Tier before / after: T4 / T4; no promotion.
No-overclaim statement: no complete production score-weighted owner, one-use $B_{(7/12)}^{\text{out}}$, production shell constants, low matrix, matching energy, strict gap, absolute anchor, either A13 gate, Nelson theorem, interacting measure, or Sector-A closure is proved.
Proof complete: false. Sector A closed: false.
Next required action: avoid global source-translation estimates. Prove a blockwise or direct fixed-law complete signed score/forest estimate that controls the replica-relative and covariance/forest residuals with source and sextic used once. Establish an aggregate one-use $B_{(7/12)}^{\text{out}}$ bound, control the near/balanced collar contribution, then apply the exact threshold $a < 2 \sqrt{(e-k^2/d)(f-k^2/d)}$ and the R-129 anchor once.

No-overclaim footer. R-133 proves exact affine and predictable-core score identities, exact score costs, a conditional polynomial-response support theorem, a precisely scoped $\gamma = 4$ coefficient-route no-go, and an exact positive- γ aggregate collar criterion. It does not prove a finite-energy production score bound, production shell constants, a strict augmented form, the absolute anchor, either A13 gate, $\text{OVERLAP}_{\text{src}}$, Nelson, removals, an interacting measure, Sector A, or any T5–T7 promotion.